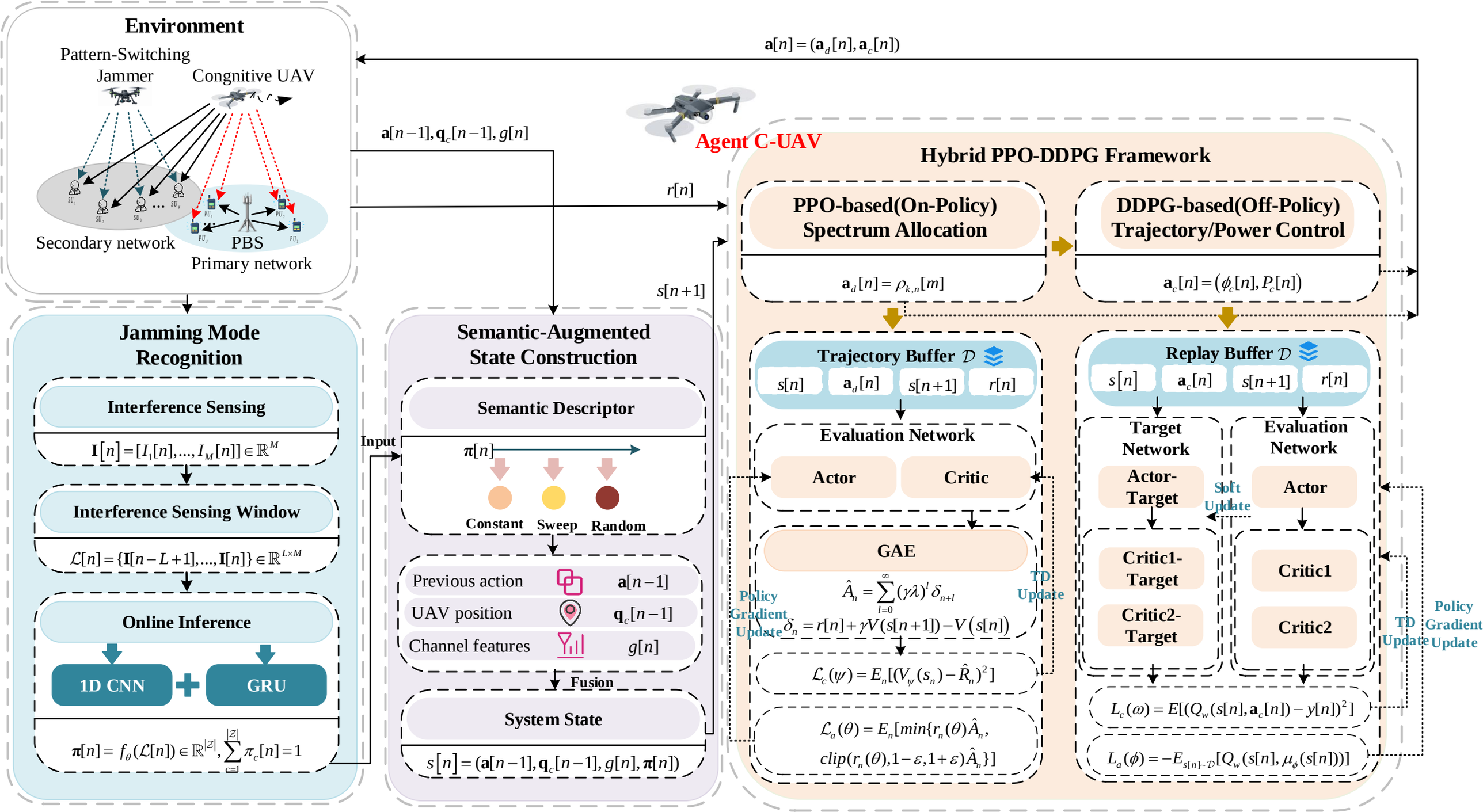
Environment
Pattern-Switching Jammer
Congnitive UAV
Secondary network
PBS
Primary network
$\mathbf{a}[n]=(\mathbf{a}_d[n],\mathbf{a}_c[n])$
$\mathbf{a}[n-1],\mathbf{q}_c[n-1],g[n]$
$r[n]$
$s[n+1]$
Agent C-UAV
Hybrid PPO-DDPG Framework
PPO-based(On-Policy) Spectrum Allocation
$\mathbf{a}_d[n]=\rho_{k,n}[m]$
DDPG-based(Off-Policy) Trajectory/Power Control
$\mathbf{a}_c[n]=(\phi_c[n],P_c[n])$
Jamming Mode Recognition
Interference Sensing
$\mathbf{I}[n]=[I_1[n],...,I_M[n]]\in\mathbb{R}^M$
Interference Sensing Window
$\mathcal{L}[n]=\{\mathbf{I}[n-L+1],...,\mathbf{I}[n]\}\in\mathbb{R}^{L\times M}$
Online Inference
1D CNN
GRU
$\boldsymbol{\pi}[n]=f_\theta(\mathcal{L}[n])\in\mathbb{R}^{|\mathcal{Z}|},\sum_{c=1}^{|\mathcal{Z}|}\pi_c[n]=1$
Input
Semantic-Augmented State Construction
Semantic Descriptor
$\boldsymbol{\pi}[n]$
Constant
Sweep
Random
Previous action $\mathbf{a}[n-1]$
UAV position $\mathbf{q}_c[n-1]$
Channel features $g[n]$
Fusion
System State
$s[n]=(\mathbf{a}[n-1],\mathbf{q}_c[n-1],g[n],\boldsymbol{\pi}[n])$
Trajectory Buffer $\mathcal{D}$
$s[n]$ $\mathbf{a}_d[n]$ $s[n+1]$ $r[n]$
Evaluation Network
Actor
Critic
GAE
$\hat{A}_n=\sum_{l=0}^{\infty}(\gamma\lambda)^l\delta_{n+l}$
$\delta_n=r[n]+\gamma V(s[n+1])-V(s[n])$
$\mathcal{L}_c(\psi)=E_n[(V_\psi(s_n)-\hat{R}_n)^2]$
$\mathcal{L}_a(\theta)=E_n[min\{r_n(\theta)\hat{A}_n,$
$clip(r_n(\theta),1-\varepsilon,1+\varepsilon)\hat{A}_n\}]$
Policy Gradient Update
TD Update
Replay Buffer $\mathcal{D}$
$s[n]$ $\mathbf{a}_c[n]$ $s[n+1]$ $r[n]$
Target Network
Actor-Target
Critic1-Target
Critic2-Target
Soft Update
Evaluation Network
Actor
Critic1
Critic2
$L_c(\omega)=E[(Q_w(s[n],\mathbf{a}_c[n])-y[n])^2]$
$L_a(\phi)=-E_{s[n]\sim\mathcal{D}}[Q_w(s[n],\mu_\phi(s[n]))]$
TD Update
Policy Gradient Update